

SQL Interview Notes

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LeetCode 1757 — Recyclable and Low Fat Products

Schema: Products

Column Name
product_id
low_fats
recyclable

Question:

Return products that are both

- low fat
- recyclable

MySQL

```
SELECT product_id
FROM Products
WHERE low_fats='Y'
AND recyclable='Y';
```

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```
SELECT product_id
FROM Products
WHERE low_fats='Y'
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```

PostgreSQL

```
SELECT product_id
FROM Products
```

```
WHERE low_fats='Y'  
AND recyclable='Y';
```

SQL Server

```
SELECT product_id  
FROM Products  
WHERE low_fats='Y'  
AND recyclable='Y';
```

2. LeetCode 584 — Find Customer Referee

Table: Customer

Column Name	Type
id	int
name	varchar
referee_id	int

Find the names of the customer that are either:
referred by any customer with id != 2.
not referred by any customer.

```
# Write your MySQL query statement below  
SELECT name  
FROM Customer  
WHERE referee_id!=2  
OR referee_id IS NULL;
```

Big Countries

Table: World

Column Name	Type
name	varchar
continent	varchar
area	int
population	int
gdp	bigint

name is the primary key (column with unique values) for this table.

A country is big if:

- it has an area of at least three million (i.e., 3000000 km2), or
- it has a population of at least twenty-five million (i.e., 25000000).

Write a solution to find the name, population, and area of the big countries.

Article Views I

Table: Views

Column Name	Type
article_id	int
author_id	int
viewer_id	int
view_date	date

There is no primary key (column with unique values) for this table, the table may have duplicate rows.

Write a solution to find all the authors that viewed at least one of their own articles.
Return the result table sorted by id in ascending order.

```
SELECT DISTINCT author_id AS id
FROM Views
WHERE author_id=viewer_id
ORDER BY id;
```

LeetCode 1683 — Invalid Tweets

Table: Tweets

Column Name	Type
tweet_id	int
content	varchar

Write a solution to find the IDs of the invalid tweets. The tweet is invalid if the number of characters used in the content of the tweet is strictly greater than 15.

```
SELECT tweet_id
FROM Tweets
WHERE LENGTH(content)>15;
```

LeetCode 627 — Swap Salary

Table: Salary

Column Name	Type
id	int
name	varchar
sex	ENUM
salary	int

The sex column is ENUM (category) value of type ('m', 'f')
Write a solution to swap all 'f' and 'm' values (i.e., change all 'f' values to 'm' and vice versa) with a single update statement and no intermediate temporary tables.

```
UPDATE Salary
SET sex=CASE
WHEN sex='m'
THEN 'f'
ELSE 'm'
END;
```

1527. Patients With a Condition

Table: Patients

Column Name	Type
patient_id	int
patient_name	varchar
conditions	varchar

patient_id is the primary key (column with unique values) for this table.

'conditions' contains 0 or more code separated by spaces

Write a solution to find the patient_id, patient_name, and conditions of the patients who have Type I Diabetes. Type I Diabetes always starts with DIAB1 prefix.

```
SELECT patient_id, patient_name, conditions
FROM Patients
WHERE conditions LIKE 'DIAB1%'
       OR conditions LIKE '% DIAB1%';
```


1873. Calculate Special Bonus

Table: Employees

Column Name	Type
employee_id	int
name	varchar
salary	int

Write a solution to calculate the bonus of each employee. The bonus of an employee is 100% of their salary if the ID of the employee is an odd number and the employee's name does not start with the character 'M'. The bonus of an employee is 0 otherwise. Return the result table ordered by employee_id.

```
SELECT employee_id,  
       CASE  
         WHEN employee_id % 2 != 0 AND name NOT LIKE 'M%' THEN salary  
         ELSE 0  
       END AS bonus  
FROM Employees  
ORDER BY employee_id;
```